# Build a complete prediction

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect net force, velocity change, and final position in one model.

**SUCCESS CRITERIA - I can...**

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain the assumptions and limits of the prediction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A 3 kg box has $\mu_s$=0.50 and $\mu_k$=0.40 on a level floor; g=9.8. Compare pushes of 6 N and 18 N, starting from rest. Find actual friction and acceleration.

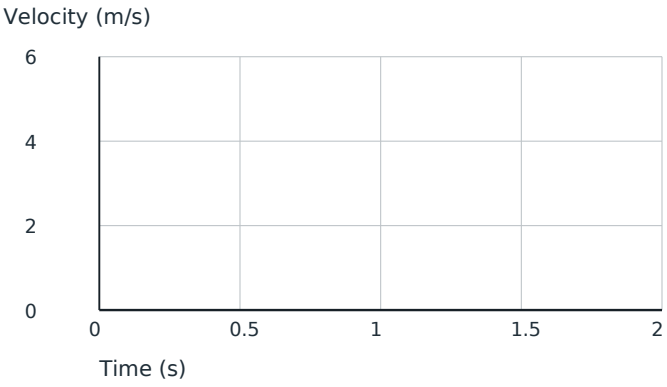
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

**Worked reasoning:** A 2 kg cart starts from rest under 6 N right and 2 N left for 2 s. Net force 4 N gives $a$=2 m/s². Its displacement is ½×2×2²=4 m right.

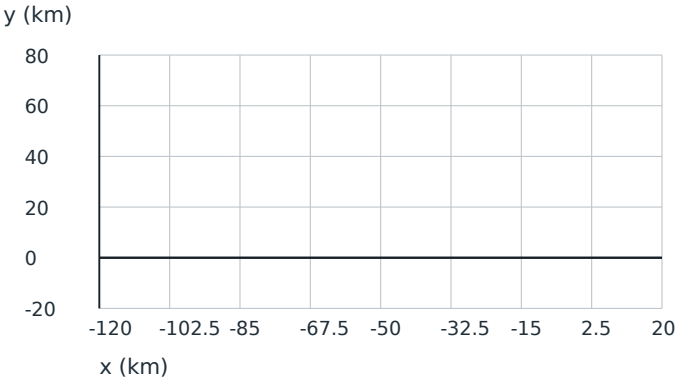
**Your turn:** Draw the force diagram and the matching velocity-time triangle. Check the 4 m using its area.




**Marzano check - circle:** ① ② ③

## 03 End point versus displacement

In a simplified tracking model, a probe starts at (-100, 0) km and moves with constant velocity (4, 3) km/s for 20 s. The origin marks a stationary reference point. Find final coordinates, displacement magnitude and final distance from the origin. Plot the route.




**Marzano check - circle:** ① ② ③

## 04 Add bounds

For the probe in question 3, the initial x coordinate has a bound of ±1 km and each velocity component has an independent bound of ±0.1 km/s. Initial y and time are exact. Find the final x and y intervals.

| Coordinate | Central (km) | Lower (km) | Upper (km) |
|---|---|---|---|
| x | ____ | ____ | ____ |
| y | ____ | ____ | ____ |

**Marzano check - circle:** ① ② ③

# Build a complete prediction

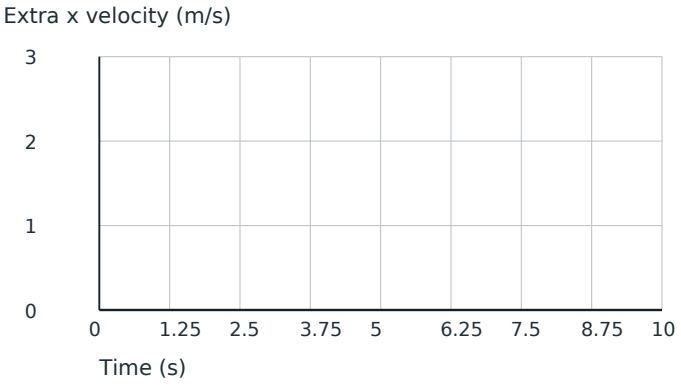
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Continue down each column.

## 05 A new force

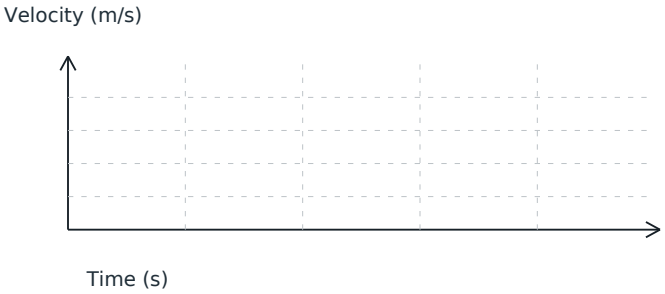
A 1000 kg probe experiences a constant 200 N net force in +x for 10 s. Compared with coasting from the same initial state, find the extra x velocity and extra x displacement. Use an acceleration-time or velocity-change graph to check.




**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A car starts at +35 m/s and brakes at -5 m/s² until rest. A student extends that acceleration to 9 s. Find the stop time/distance and explain why extension is invalid for ordinary braking.
Sketch the velocity-time model; label the given values.




**Marzano check - circle:** ① ② ③

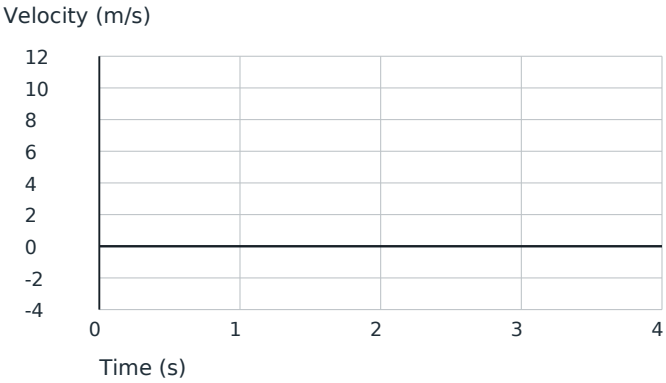
## 07 Exit check

Write a short public explanation of questions 3 and 4. Include the predicted position and coordinate ranges. Explain two reasons this simplified model cannot establish a real asteroid collision forecast.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

A cart starts at x=3 m with v=-2 m/s. Its mass is 2 kg and constant net force is +6 N for 4 s. Predict final velocity and position. Check whether it turns.




**Marzano check - circle:** ① ② ③